

Design Statement

Simulating the Amazon Homepage Layout - 03CIT4053 Technology for Web Development

1. Student Information

Item	Details
Student ID	25525116
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Course	03CIT4053 Technology for Web Development
Instructor	Liam XU
GitHub Repository (public)	https://github.com/poucaiije000/25525116_TsoiKit.git

2. Page Structure

The homepage is divided into six blocks, each with one clear purpose. I split the page this way so that every block can be built and tested independently, and each commit in the Git history maps to one block.

#	Block	Main class	How it is laid out
1	Top navigation bar	.nav-belt	logo float:left, search box float:left, right group float:right
2	Secondary nav bar	.nav-main	category links .sub-item float:left
3	Promo banner row	.banner-row	5 banner cards float:left, width 20%
4	Product card area	.card-row	two rows, 4 white cards float:left per row
5	Sign-in recommendation	.signin-row	normal flow, text centered, button margin:0 auto
6	Footer	.footer	4 link columns float:left + copyright bar

3. Box Model Usage

All widths are calculated with the content-box model taught in the course (width only covers the content area; padding and border are added outside). Every number below was calculated on paper before writing the CSS, which kept the cards in one row without overflow.

Calculation	Result
Card: width 295px + padding 20px x2	actual box = 335px
One card row: 4 x 335px + 3 x 20px gaps	= 1400px container width
Card inner width: 295 - 40 (padding)	= 255px
2x2 image grid: 118 x2 + 9px gap	= 245px <= 255px, fits one line
Footer: 4 x 220px columns + 3 x 40px gaps	= 1000px
Banner: 5 x 20%	= 100% of 1400px row

margin: 0 auto is used to center fixed-width block boxes: .banner-row, .card-row, .footer-inner, .signin-btn and .card-imgs (image grid).

4. Normal Flow vs Float

Rule of thumb I followed: keep the vertical rhythm in normal flow, use float only when elements must sit side by side on one line.

Technique	Used for	Why
Normal flow	nav texts, card titles, sign-in row, footer headings/links	content stacks top-to-bottom; float is not needed
float: left	logo, search box, right group (top nav)	side-by-side layout on the same bar
float: left	sub nav links, banner cards, product cards, footer columns	horizontal arrangement of block-level elements
float: left	4 small images inside each card	2x2 grid inside the card

5. Clearing Floats (Parent Height Collapse)

When children are floated, the parent no longer stretches to contain them (height collapse). I met this problem and fixed it with three different methods from the lecture notes.

Method	Where used	How
overflow: hidden	.nav-belt-inner (top nav)	parent gets a new block context and wraps the floats
extra empty tag	.nav-main-inner, .banner-row, .footer-inner	a <code><div class="clear"></div></code> with <code>clear:both</code> at the end
::after pseudo-element	.card-row	<code>content:""; display:block; clear:both</code> on the parent

6. Debugging Process

Problem 1 - image grid overflow. The 2x2 grid of small images was pushed to a new line. I calculated the real space: $140\text{px} \times 2 + 9\text{px} \times 2 = 298\text{px}$, which is wider than the 295px inner width, so the last image wrapped. After switching to the content-box model, the inner width became 255px, so I resized the images to 118px ($118 \times 2 + 9 \times 2 = 254\text{px} \leq 255\text{px}$).

Problem 2 - box-sizing. I initially used box-sizing: border-box to keep a 335px card. Checking the lecture notes, only box-sizing: content-box appears there (the classic box model for teaching), so I removed border-box and recalculated with content-box to stay within what was taught.

Problem 3 - grid not centered. The 245px grid left 10px of empty space on the right of the card. I centered it with margin: 0 auto on a fixed-width .card-imgs container, using the same centering technique as the card row.

Problem 4 - different card sizes. Cards looked uneven because titles wrapped to two lines on some cards. I gave .card a fixed height and .card h3 a fixed 44px height with overflow: hidden, so the image grid starts at the same level on every card.

7. Reflection

The most useful habit I built was calculating widths before writing CSS instead of guessing and adjusting. Writing down the numbers (335px card, 1400px row, 245px grid) made layout predictable and the code shorter. I also understand now why float needs to be cleared: without it, the parent collapses and the next block slides up under the floating elements.

8. AI Usage Declaration

AI tools were used as learning support: I asked about the overall framework of the webpage, how to adjust widths and spacing, and CSS concepts (box model, float, clearing floats). AI was also used as a final check of the code. All HTML and CSS code was written and committed by me, step by step, as shown in the Git history below.

9. Screenshots

9.1 Final page layout (top of the page)

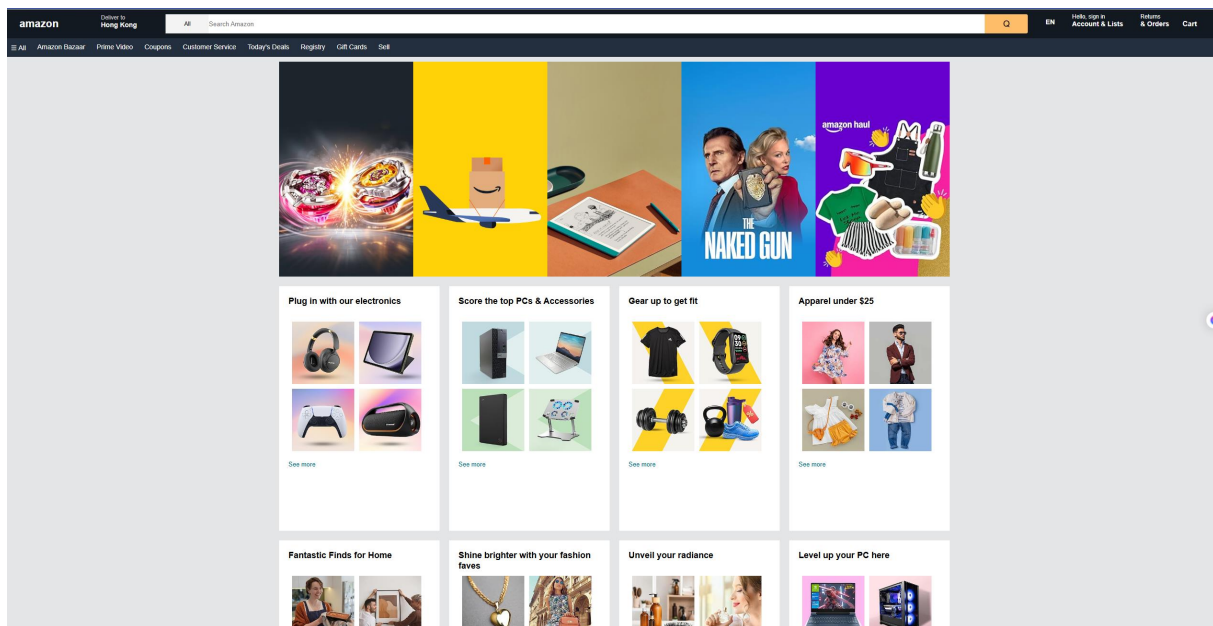


Figure 1. Top of the final page: top nav, sub nav, banner row and the first card row.

9.2 Final page layout (bottom of the page)

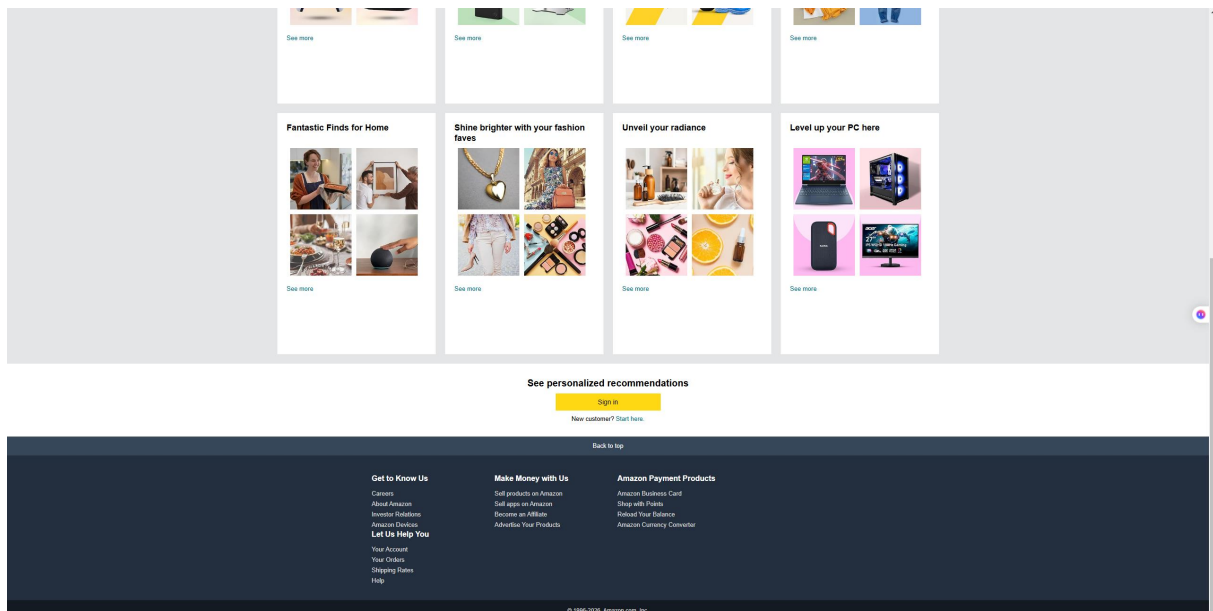


Figure 2. Bottom of the final page: two card rows, sign-in recommendation and the footer.

9.3 Git commit history

Commits

main

All usersAll time

Commits on Oct 1, 2026

polish code: add comments and fix card layout
poucaijie000 committed 7 minutes ago
b62a786

add sign-in recommendation and footer
poucaijie000 committed 1 hour ago
2d46411

Commits on Sep 29, 2026

add two card rows with image grid
poucaijie000 committed 2 days ago
00e075

add promo banner row
poucaijie000 committed 2 days ago
28e990c

Commits on Sep 28, 2026

add sub nav links and clear float with extra tag
poucaijie000 committed 3 days ago
716f7b0

Commits on Sep 27, 2026

add top nav with logo search and cart
poucaijie000 committed 4 days ago
bf49deb

init: folder structure and index.html skeleton
poucaijie000 committed 4 days ago
bcb99e6

Figure 3. Seven commits pushed over four days (Sep 27 - Oct 1).